

ECE253/CSE208 Introduction to Information Theory

Lecture 2: Probability Theory Revisited

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- H. Stark and J. Woods, "Probability, Random Processes, and Estimation Theory for Engineers".
- A. Papoulis, "Probability, Random Variables, and Stochastic Processes".
- P. Cameron, "Notes on Probability".

Why Probability Theory for Information Theory?

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CDF/PMF/PDF

Expectation and
Variance

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RV Convergence

Concentration
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In this lecture, we revisit probability concepts that will be used throughout information theory:

- Random variables and distributions $\rightarrow$ source and channel models
- Expectation and variance $\rightarrow$ average distortion, power constraints
- Indicator functions $\rightarrow$ typical sets and empirical frequencies
- Conditional probability $\rightarrow$ decoding and inference
- LLN and concentration $\rightarrow$ reliability of long block codes
- Gaussian distributions $\rightarrow$ optimal signaling and Gaussian channels

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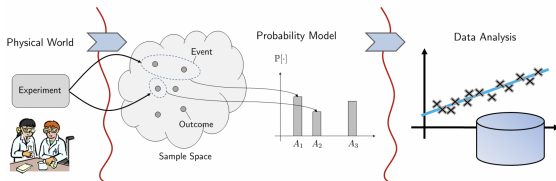


Figure: source: Prof. Stanley Chan at Purdue Univ.

- **Physical world** $\rightarrow$ **sample space** Ω : A real-world experiment is abstracted into a sample space whose elements represent all possible outcomes.
- **Outcomes** $\rightarrow$ **events** $\rightarrow$ **probabilities**: Events are subsets of Ω , and a probability measure $P(\cdot)$ assigns probabilities to these events.
- **Probability model** $\rightarrow$ **data analysis**: The probabilistic model provides the foundation for statistical inference, learning, and data-driven decision making.

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Probability space is a three-tuple $(\Omega, \mathcal{F}, P)$:

- **Sample space** Ω : the set of all outcomes of a random experiment. Ω may be finite ($\{H, T\}$), countably infinite ($\Omega = \{1, 2, 3, \dots\}$), or uncountably infinite ($[0, 1]$).
- **Event space** $\mathcal{F}$ is a collection of subsets of Ω to which probabilities are assigned. Any element $A \in \mathcal{F}$ is called an event.
- **Probability function** P : satisfies three axioms:
 - $P(A) \geq 0, \forall A \in \mathcal{F}$
 - $P(\Omega) = 1$
 - If $A_1, A_2, \dots$ are mutually exclusive events; i.e. $A_i \cap A_j = \emptyset, \forall i \neq j$, then

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$$

Basic Properties of Probability

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Consider events $A, B, C \subseteq \Omega$. Let $P(AB)$ denote $P(A \cap B)$.

- $A \subseteq B \Rightarrow P(A) \leq P(B)$.
- $P(AB) \leq \min\{P(A), P(B)\}$.
- $P(A \cup B) = P(A) + P(B) - P(AB)$.
- $P(\bar{A}) = 1 - P(A)$ where $\bar{A} = \Omega \setminus A$ is the complement of A .
- De Morgan's law: $\overline{AB} = \bar{A} \cup \bar{B}$; $\overline{A \cup B} = \bar{A} \bar{B}$
- Distributive law: $(A \cup B) \cap C = (AC) \cup (BC)$.

Venn Diagram

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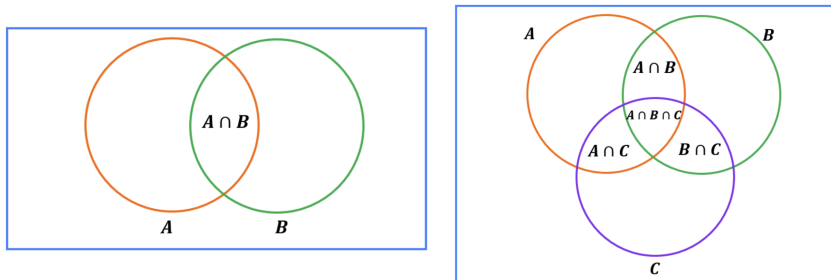


Figure: A Venn diagram is a diagram that shows the relationship between and among a finite collection of sets.

Conditional Probability and Bayes' Rule

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- **Conditional probability:** $P(A|B) := \frac{P(AB)}{P(B)}$ if $P(B) > 0$.
- $P(AB) = P(A|B)P(B) = P(B|A)P(A)$.
- If A and B are independent, then $P(AB) = P(A)P(B)$.

Bayes' Rule: Posterior = Prior $\times$ Data consistency (renormalized)

$$P(H | D) = \frac{P(D | H) P(H)}{P(D)}$$

- **Prior** $P(H)$: belief about the hypothesis before observing data.
- **Likelihood** $P(D | H)$: how likely the data is if the hypothesis is true.
- **Evidence** $P(D)$: normalizing constant over all hypotheses.
- **Posterior** $P(H | D)$: updated belief after observing data.

Conditional vis-à-vis Unconditional Probability

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Q: $P(A|B) \stackrel{<}{\geq} P(A)$?

A: It can be any case in general.

Some special cases:

- If A and B are independent (denoted as $A \perp B$), then $P(A|B) = P(A)$.
- If $AB = \emptyset$ and $0 < P(A), P(B) < 1$, then
$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(\emptyset)}{P(B)} = 0 < P(A) \Rightarrow P(A|B) < P(A).$$
- If $B \subset A$ and $P(A) < 1$, then
$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(B)}{P(B)} = 1 > P(A) \Rightarrow P(A|B) > P(A).$$
- If $A \subseteq B$, and $0 < P(B) < 1$, then
$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} > P(A).$$

A Concrete Example

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Take a standard deck of cards (52 cards without Jokers). Remove all black queens and kings. When picking a card, define 3 events:

- A : The card picked is a face card.
- B_1 : The card picked is a heart.
- B_2 : The card picked is a spade.

Question: Find the values of $P(A)$, $P(A|B_1)$, $P(A|B_2)$.

- A : 48 cards left with 8 face cards: $P(A) = \frac{1}{6}$.
- B_1 : 13 cards of hearts left with 3 face cards: $P(A|B_1) = \frac{3}{13}$.
- B_2 : 11 card of spades left with only 1 face card (the Jack of Spades): $P(A|B_2) = \frac{1}{11}$.

We have $P(A|B_2) < P(A) < P(A|B_1)$.

Monty Hall Problem¹

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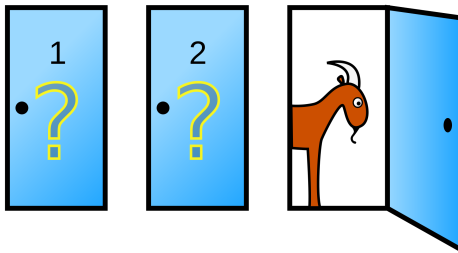


Figure: Suppose you are on a game show with three doors. One hides a car and the other two hide goats. Suppose you choose Door 1. The host, who knows the contents, opens a goat door you did not pick (say Door 3). Then, you are offered the option to switch to Door 2. Is switching better?

¹https://en.wikipedia.org/wiki/Monty_Hall_problem

Why Switching Is Better (1/2)

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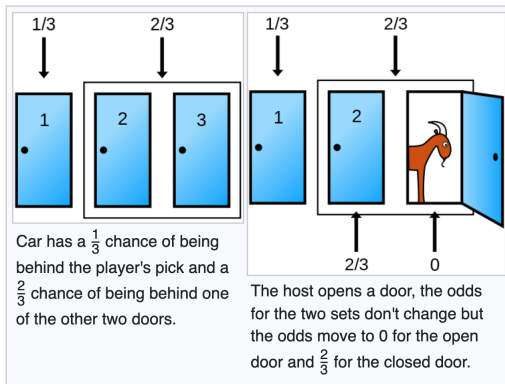
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- Alternatively, think of the complement. The only way to get it wrong by switching is to have picked the correct door in the first place. The odds of picking the correct door first are $\frac{1}{3}$.
- **The more you know, the better your decision!**

Why Switching Is Better (2/2)

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“The host opens Door 3 and it is a goat”: This describes what happened in a particular run of the game. But across all possible underlying car locations, the host might open Door 2 or Door 3 depending on where the car is.

Door 1	Door 2	Door 3	Stay	Switch
Goat	Goat	Car	Goat	Car
Goat	Car	Goat	Goat	Car
Car	Goat	Goat	Car	Goat

- Staying wins in 1 out of 3 cases. Switching wins in 2 out of 3 cases.
- The host's action is **not random**: it depends on your initial choice².
- The remaining unopened doors are **not equally likely**.

²Key subtlety: Door 3 is not fixed to be a goat “in advance”.

Chain Rule for Conditional Probability

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For 3 events, A , B and C , we have

$$P(ABC) = P(C|AB)P(AB) = P(C|AB)P(B|A)P(A).$$

This can be generalized to n events $A_1, A_2, \dots, A_n$ (let $A_0 := \Omega$). We have

$$\begin{aligned} P(A_1 A_2 \cdots A_n) &= \prod_{i=1}^n P(A_i | A_1 A_2 \cdots A_{i-1}) \\ &= P(A_n | A_1 A_2 \cdots A_{n-1}) \times P(A_{n-1} | A_1 A_2 \cdots A_{n-2}) \times \cdots \\ &\quad \times P(A_2 | A_1) \times P(A_1) \end{aligned}$$

Partition Theorem

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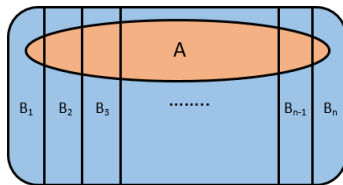
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Theorem (Partition Theorem, a.k.a. Law of Total Probability)

Let $B_1, B_2, \dots, B_n$ form a **partition** (i.e., $B_i B_j = \emptyset, \forall i \neq j$ and $\bigcup_i B_i = \Omega$) of the sample space Ω , and assume $P(B_i) \neq 0$ for all i . Then,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$$



The Birthday Paradox (1/3)

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Given n people named $p_1, p_2, \dots, p_n$, what is the probability that at least two of them have the same birthday?

Solution 1:

Let A_2 be the event that p_2 has a different birthday from p_1 . p_1 has their birthday on one of the days of the entire year. Hence, $P(A_2) = 1 - \frac{1}{365}$.

Let A_3 be the event that p_3 has a different birthday from p_2 and p_1 . So, $A_3|A_2$ denotes the event that p_3 has a different birthday from p_2 and p_1 given that p_2 and p_1 have different birthdays. We have $P(A_3|A_2) = 1 - \frac{2}{365}$.

The Birthday Paradox (2/3)

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A_2A_3 is the event that p_1, p_2 , and p_3 have 3 different birthdays.

$$P(A_2A_3) = P(A_3|A_2)P(A_2) = \left(1 - \frac{2}{365}\right) \left(1 - \frac{1}{365}\right) = \frac{363 \times 364}{365^2} \approx 0.9918$$

$$\begin{aligned} P(A_3) &= P(A_3|A_2)P(A_2) + P(A_3|\bar{A}_2)P(\bar{A}_2) \\ &= \left(1 - \frac{2}{365}\right) \left(1 - \frac{1}{365}\right) + \left(1 - \frac{1}{365}\right) \times \frac{1}{365} \approx 0.9945 \end{aligned}$$

Now, define a general A_i as the event that the birthday of p_i is not the same day as any of the birthdays of $p_1, p_2, \dots, p_{i-1}$. We have $P(A_i|A_1A_2 \cdots A_{i-1}) = 1 - \frac{i-1}{365}$.

The Birthday Paradox (3/3)

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The probability that all n people have different birthdays is

$$\begin{aligned} q_n &:= P(A_1 A_2 \cdots A_n) = \prod_{i=1}^n P(A_i | A_1 A_2 \cdots A_{i-1}) \\ &= \left(1 - \frac{n-1}{365}\right) \times \left(1 - \frac{n-2}{365}\right) \times \cdots \times \left(1 - \frac{1}{365}\right) \end{aligned}$$

Note that $P(A_1) = 1 - \frac{0}{365} = 1$.

- the complement of $A_1 \cdots A_n$ is: at least two people have the same birthday.
- the required probability is $b_n = 1 - q_n$,
- b_n is increasing in n : $b_{23} \approx 0.507$; $b_{30} \approx 0.706$; $b_{40} \approx 0.891$; $b_{70} \approx 0.999$.

Solution 2:

The total number of all possibilities is 365^n . The number of cases that n people all have different birthdays is 365 permute n : ${}^{365}P_n = \frac{365!}{(365-n)!}$. Hence, $q_n = \frac{365!}{(365-n)! \times 365^n}$.

Random Variables (1/2)

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Definition (Random Variable (RV))

Let Ω be the sample space of an experiment, and $\mathbb{R}$ denote the set of real numbers. Then, a *random variable* $X : \Omega \mapsto \mathbb{R}$ associated with this experiment is a function that assigns each outcome in Ω to a real number. The support of X is denoted as $\text{supp}(X)$.

Example. Flip a coin 5 times. Let X denote the random variable for the number of times the coin came up heads. Then $X(\omega_0) = 3$, for the outcome $\omega_0 = \{\text{HHTHT}\}$.

Two different types of random variables that are often studied: **discrete** and **continuous**.

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If X is a discrete random variable (when $\text{supp}(X)$ is a countable set), we use the notation

$$\Pr(X = k) := \Pr(\{\omega : X(\omega) = k\})$$

for the probability of the event $X = k$.

If X is a continuous random variable (when $\text{supp}(X)$ is an uncountable set), we use the notation

$$\Pr(a \leq X \leq b) := \Pr(\{\omega : a \leq X(\omega) \leq b\})$$

for the probability of the event $a \leq X \leq b$.

Cumulative Distribution Function (CDF)

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Definition

Let X be a random variable associated with an experiment. Then,

$$F_X(x) := \Pr(X \leq x)$$

is a *cumulative distribution function*.

Properties of CDFs:

- (a) $0 \leq F_X(x) \leq 1$.
- (b) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
- (c) F_X is nondecreasing, namely if $x \leq y$ then $F_X(x) \leq F_X(y)$.
- (d) F_X is right-continuous, i.e. $\lim_{x \rightarrow a^+} F_X(x) = F_X(a)$.

CDF for Discrete and Continuous RVs

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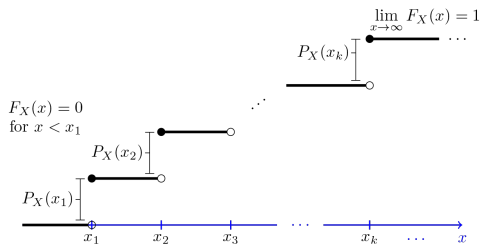


Figure: CDF of a discrete RV: staircase function.

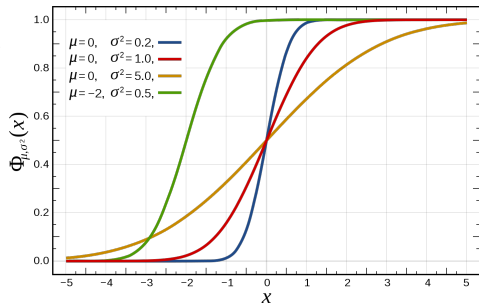


Figure: CDF of a continuous RV: continuous function.

PMF and PDF

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For discrete random variables, we have the *probability mass function* (PMF):

$$p_X(x) := \Pr(X = x), \text{ where } \sum_{x \in \text{supp}(X)} p_X(x) = 1.$$

For continuous random variables, we instead consider *probability density function* (PDF),

$$f_X(x) := \frac{dF_X(x)}{dx} = F'_X(x),$$

if F_X is differentiable at x .³ Notice that $f_X(x)$ and $\Pr(X = x)$ are two different concepts, which can be related by

$$\Pr(x \leq X \leq x + \Delta x) \approx f_X(x) \Delta x$$

and

$$\Pr(X \in A) = \int_{x \in A} f_X(x) dx.$$

³Note that F_X may not be everywhere differentiable even for continuous RV.

Bernoulli Distribution

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Definition (Models a single binary trial with success probability p)

A Bernoulli random variable $X \sim \text{Bern}(p)$ takes values in $\{0, 1\}$, with PMF

$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad 0 \leq p \leq 1.$$

Or, more compactly

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Key Properties

- Finite support: $\{0, 1\}$; $\mathbb{E}[X] = p$ and $\text{Var}(X) = p(1 - p)$
- Let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(p)$. Then $\sum_{i=1}^n X_i \sim \text{Binomial}(n, p)$.

Poisson Distribution

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Definition (Models the number of events in a fixed interval with rate λ)

A Poisson random variable $Y \sim \text{Pois}(\lambda)$ takes values in $\{0, 1, 2, \dots\}$, with PMF

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots, \lambda > 0.$$

Key Properties

- Countably infinite support: $\{0, 1, 2, \dots\}$; $\mathbb{E}[Y] = \lambda$ and $\text{Var}(Y) = \lambda$
- Additivity: If $Y_1 \sim \text{Pois}(\lambda_1)$ and $Y_2 \sim \text{Pois}(\lambda_2)$ are independent, then $Y_1 + Y_2 \sim \text{Pois}(\lambda_1 + \lambda_2)$
- Limit of Binomial(n, p) as $n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$

Univariate Gaussian (Normal) Distribution

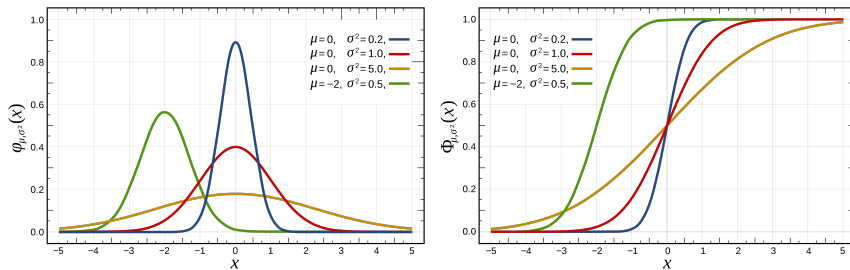


Figure: PDF and CDF of Gaussian distribution.

PDF of Gaussian distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

CDF of standard Gaussian distribution ($\mu = 0, \sigma = 1$): $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$

Common Distributions

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Name of the probability distribution	Probability distribution function	Mean	Variance
Binomial distribution	$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Geometric distribution	$\Pr(X = k) = (1 - p)^{k-1} p$	$\frac{1}{p}$	$\frac{(1 - p)}{p^2}$
Normal distribution	$f(x \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Uniform distribution (continuous)	$f(x a, b) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
Exponential distribution	$f(x \lambda) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Figure: PDF or PMF of commonly used random variables (Wiki).

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Definition (Expectation)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function and X be a RV, discrete or continuous, then the *expectation* of the random variable $g(X)$ is

$$\mathbb{E}[g(X)] := \sum_{x \in \text{supp}(X)} g(x)p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] := \int_{x \in \text{supp}(X)} g(x)f_X(x)dx.$$

Linearity of the expectation operator:

- (a) $\mathbb{E}[ag(X) + bh(X)] = a\mathbb{E}[g(X)] + b\mathbb{E}[h(X)]$ for any constants a, b , and arbitrary functions $g(\cdot), h(\cdot)$.
- (b) $X \perp Y \Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Variance

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Given a distribution of a random variable, we use the notion of variance to measure how concentrated that distribution is around the expectation (mean). Formally, we have

Definition (Variance)

$\text{Var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, where $\mathbb{E}[X^2]$ is the second moment of X .

The following can be derived immediately from the definition:

- (a) $\text{Var}[cX] = c^2 \text{Var}[X]$.
- (b) $\text{Var}[c] = 0$ for any constant c .
- (c) $\text{Var}[aX \pm bY] = a^2 \text{Var}[X] + b^2 \text{Var}[Y] \pm 2ab \times \text{Cov}[X, Y]$.

Example: Expectation and Variance of an Exponential RV

Let $X \sim \exp(\lambda)$ with density $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$. Find $\mathbb{E}[X]$ and $\text{Var}[X]$.

Expectation:

$$\mathbb{E}[X] = \int_0^{\infty} x f_X(x) dx = \lambda \int_0^{\infty} x e^{-\lambda x} dx$$

Integration by parts gives

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

Variance:

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx - \frac{1}{\lambda^2}$$

Using integration by parts twice,

$$\text{Var}[X] = \frac{1}{\lambda^2}.$$

Indicator Function

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The *indicator function*:

$$\mathbf{1}_A := \begin{cases} 1, & \text{if } A \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

- $\mathbb{E}[\mathbf{1}_A] = \Pr(A) \times 1 + \Pr(\bar{A}) \times 0 = \Pr(A).$
- $F_X(x) = \mathbb{E}[\mathbf{1}_{\{X \leq x\}}].$
- **Counting via indicators:**

$$\sum_{i=1}^n \mathbf{1}_{\{X_i \in A\}} \quad \text{counts how many samples fall in } A.$$

- $\mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_{\{X_i \in A\}}\right] = n \Pr(X \in A).$
- **Typicality:** $\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i = x\}} \approx P_X(x)$ for large n .

Note: This idea underlies empirical frequencies, typical sets, and many arguments in info theory.

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Definition (Joint CDF)

Let X, Y be two RVs associated with an experiment. Then the *joint cdf* of X and Y is $F_{XY} := \Pr(X \leq x, Y \leq y)$ and the *marginal cdf* of X is $F_X(x) := \lim_{y \rightarrow +\infty} \Pr(X \leq x, Y \leq y)$.

- *joint PMF* $p_{XY}(x, y) := \Pr(X = x, Y = y)$ and *joint PDF* $f_{XY}(x, y) := \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$

Marginalization:

- *marginal PMF* of X $p_X(x) = \sum_{y \in \text{supp}(Y)} \Pr(X = x, Y = y)$.
- *marginal PDF* of X $f_X(x) = \int_{y \in \text{supp}(Y)} f_{XY}(x, y) dy$.
- $\iint_A f_{XY}(x, y) dx dy = \Pr((x, y) \in A)$.

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Definition (Conditional Distribution for Discrete RVs)

Let X and Y be discrete RVs. The *conditional distribution* of Y given $X = x$ is defined as

$$p_{Y|X}(y | x) := \frac{p_{XY}(x, y)}{p_X(x)}, \quad \text{for } p_X(x) > 0.$$

Definition (Conditional Distribution for Continuous RVs)

Let X and Y be continuous RVs. The conditional distribution (density) of Y given $X = x$ is defined as

$$f_{Y|X}(y | x) := \frac{f_{X,Y}(x, y)}{f_X(x)}, \quad \text{for } f_X(x) > 0.$$

Independence

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Definition (Independent RVs)

X and Y are independent, denoted as $X \perp Y$, if

$$p_{Y|X}(y | x) = p_Y(y) \iff p_{XY}(x, y) = p_X(x)p_Y(y).$$

Or,

$$f_{Y|X}(y | x) = f_Y(y) \iff f_{X,Y}(x, y) = f_X(x) f_Y(y).$$

Lemma

If $X \perp Y$, then $g(X) \perp h(Y)$ holds for any measurable functions $g(\cdot)$ and $h(\cdot)$.

Interpretation: independent sources remain independent under separate processing.

Covariance & Correlation (1/2)

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Definition (Covariance and Correlation)

Let X and Y be two random variables. Their *covariance* is defined as

$$\text{Cov}[X, Y] := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Their *correlation* is defined as $\text{Corr}[X, Y] := \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$

Covariance & Correlation (2/2)

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1. $\text{Corr}[X, Y] \in [-1, 1]$ is a measure of *linear association* between X and Y .
2. X and Y are uncorrelated if $\text{Corr}[X, Y] = 0$.
3. $\text{Corr}[X, Y] = \pm 1 \Leftrightarrow Y = aX + b$ for some constants a and b .
4. If $X \perp Y \Rightarrow$ they are uncorrelated.
5. X and Y can be uncorrelated yet dependent due to a nonlinear relationship.

Example: $X \sim N(0, 1), Y = X^2 \Rightarrow \text{Corr}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3] = 0$.
Hence, X and Y are uncorrelated. But they are clearly dependent.

Example 1 of Covariance

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Let X and Y be discrete random variables with joint probability mass function

$$p_{X,Y}(x,y) = \begin{cases} \frac{1}{2}, & x = 3, y = 4, \\ \frac{1}{3}, & x = 3, y = 6, \\ \frac{1}{6}, & x = 5, y = 6, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the expectations:

$$\mathbb{E}[X] = 3 \times \left(\frac{1}{2} + \frac{1}{3}\right) + 5 \times \left(\frac{1}{6}\right) = \frac{10}{3}, \quad \mathbb{E}[Y] = 4 \times \left(\frac{1}{2}\right) + 6 \times \left(\frac{1}{3} + \frac{1}{6}\right) = 5.$$

Hence, the covariance is

$$\begin{aligned} \text{Cov}(X,Y) &= \mathbb{E}\left[\left(X - \frac{10}{3}\right)(Y - 5)\right] \\ &= \frac{1}{3}. \end{aligned}$$

Example 2 of Covariance

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Let X be any random variable with $\text{Var}(X) > 0$.

Define $Y = 3X$, $Z = -4X$, then $\mu_Y = 3\mu_X$, $\mu_Z = -4\mu_X$.

We have

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[(X - \mu_X)(3X - 3\mu_X)] \\ &= 3 \mathbb{E}[(X - \mu_X)^2] = 3 \text{Var}(X).\end{aligned}$$

Similarly,

$$\text{Cov}(X, Z) = \mathbb{E}[(X - \mu_X)(-4X + 4\mu_X)] = -4 \text{Var}(X).$$

Interpretation: $\text{Cov}(X, Y) > 0$ since Y increases with X , while $\text{Cov}(X, Z) < 0$ since Z decreases with X .

Multivariate Gaussian Distribution

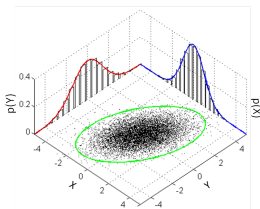


Figure: Elliptical level sets:
The density contours are ellipsoids centered at μ , with shape determined by Σ .
(from Wiki).

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, the PDF for multivariate Gaussian distribution is:

$$f(\mathbf{x}) := \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu) \right)$$

where the mean vector $\mu = [\mu_1, \dots, \mu_n]^\top \in \mathbb{R}^n$ and the covariance matrix Σ is an $n \times n$ positive semidefinite matrix. Its (i, j) -entry is $\text{Cov}[X_i, X_j]$. The density is completely determined by μ and Σ .

Maximum entropy: Among all distributions with fixed covariance, the Gaussian distribution has the largest differential entropy.

Key Properties of Multivariate Gaussian Distribution

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- **Scalar projection:** A random vector $\mathbf{X} \in \mathbb{R}^n$ is multivariate Gaussian if and only if every scalar projection $\mathbf{t}^\top \mathbf{X}$ is a univariate Gaussian random variable for all $\mathbf{t} \in \mathbb{R}^n$.
- **Gaussian marginals:** Any subvector of a multivariate Gaussian random vector is itself Gaussian. The converse is not true.
- **Gaussian conditionals:** Conditional distributions of jointly Gaussian random vectors are also Gaussian.
- **Uncorrelated $\Leftrightarrow$ independent (Gaussian only):** For jointly Gaussian random variables, zero covariance implies independence.
- **Linear transformations:** Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. For any matrix $\mathbf{A}$ and vector $\mathbf{b}$,

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

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Theorem (Weak Law of Large Numbers (WLLN))

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) RVs with $\mu = \mathbb{E}[X_1]$ and $\text{Var}(X_1) < \infty$.

Define the sample mean

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$, $\bar{X}_n \xrightarrow{P} \mu$, i.e., for every $\varepsilon > 0$, $\Pr(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$.

Information-theoretic relevance

LLN explains why empirical averages converge to expected values, which underlies typical sequences and asymptotic equipartition.

CLT

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Theorem (Central Limit Theorem (CLT))

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. RVs, and assume that $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, for all i . Let $S_n := \sum_{i=1}^n X_i$. Then, $\mathbb{E}[S_n] = n\mu$, $\text{Var}[S_n] = n\sigma^2$ and we have the standardization of S_n :

$$Z_n := \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{\text{dist}} N(0, 1) \text{ as } n \rightarrow \infty$$

where $N(0, 1)$ is the standard normal distribution.

Information-theoretic relevance

CLT motivates Gaussian approximations and second-order (finite-blocklength) analysis.

Convergence of Random Variables

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Definition (Convergence of random variables)

The given sequence of random variables $X_1, X_2, \dots$ converges to a random variable X :

1. Almost sure ($X_n \xrightarrow{\text{a.s.}} X$) [a.k.a. convergence with probability 1] if $\Pr \{\lim_{n \rightarrow \infty} X_n = X\} = 1$
2. In probability ($X_n \xrightarrow{P} X$) if for every $\epsilon > 0$, $\lim_{n \rightarrow \infty} \Pr \{|X_n - X| > \epsilon\} = 0$
3. In distribution ($X_n \xrightarrow{\text{dist}} X$) if $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$ for all continuity points x of $F_X(x)$
4. In r^{th} -order mean ($X_n \xrightarrow{L^r} X$) if $\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^r] = 0$
5. In mean square (special case when $r = 2$) if $\lim_{n \rightarrow \infty} \mathbb{E}[(X_n - X)^2] = 0$

Note: Almost sure, in probability, and L^r convergence require a common probability space; convergence in distribution does not (comparing only laws).

Pathwise Comparison of Convergence Notions (1/3)

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Let $(\Omega, \mathcal{F}, P)$ be a fixed probability space, and let $X_n, X : \Omega \rightarrow \mathbb{R}$.

Pointwise convergence (everywhere)

$$X_n(\omega) \rightarrow X(\omega) \quad \text{for every } \omega \in \Omega.$$

No exceptions are allowed, including outcomes in sets of probability zero.

Pathwise Comparison of Convergence Notions (2/3)

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Almost sure (a.s.) convergence

$$X_n \xrightarrow{\text{a.s.}} X \iff \Pr\left(\left\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

Equivalently, there exists $\Omega_0 \subseteq \Omega$ with $P(\Omega_0) = 1$ such that

$$\forall \omega \in \Omega_0 : X_n(\omega) \rightarrow X(\omega).$$

A null set of “bad” outcomes $(\Omega \setminus \Omega_0)$ is permitted.

- Almost sure convergence (informal): With probability 1, the outcomes eventually settle down and stay close to the target forever.

Pathwise Comparison of Convergence Notions (3/3)

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Convergence in probability (i.p.)

$$X_n \xrightarrow{P} X \iff \forall \varepsilon > 0 : \Pr(\{\omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) \xrightarrow{n \rightarrow \infty} 0.$$

For each n , some outcomes may violate the tolerance, but their total probability vanishes.

- Convergence in probability (informal): As you repeat an experiment more and more times, the chance of being far from the target value becomes very small.

Strong vs Weak Convergence

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1. Strong convergence: pointwise everywhere (p.e), almost sure (a.s.), and in L^r .
2. Weak convergence: in probability (i.p.) and in distribution (i.d.).

Hierarchy (strong $\rightarrow$ weak):

- p.e. $\Rightarrow$ a.s. $\Rightarrow$ i.p. $\Rightarrow$ i.d.
- $L^r \Rightarrow$ i.p. $\Rightarrow$ i.d.
- p.e. $\not\Rightarrow L^r$; a.s. $\not\Rightarrow L^r$

Concentration Inequalities

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Concentration inequalities provide bounds on how a random variable deviates from some value (e.g., its expected value):

- Markov's inequality: If X is a nonnegative RV, then $\Pr(X \geq t) \leq \frac{\mathbb{E}(X)}{t}$, for any $t > 0$.
- Chernoff's inequality: If X is a nonnegative RV, then $\Pr(X \geq t) = \Pr(e^{aX} \geq e^{at}) \leq \frac{\mathbb{E}(e^{aX})}{e^{at}}$, for any $a > 0$.
- Chebyshev's inequality: $\Pr(|X - \mathbb{E}(X)| \geq t) \leq \frac{\text{Var}(X)}{t^2}$, for any $t > 0$.
- Hoeffding's inequality: Consider the empirical mean $\bar{X}_n := \frac{1}{n}(X_1 + \cdots + X_n)$ for independent random variables $X_i \in [a_i, b_i]$ for all i . Then, $\Pr(|\bar{X}_n - \mathbb{E}(\bar{X}_n)| \geq t) \leq 2 \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$, for any $t > 0$.

Relationships among Common Discrete Distributions

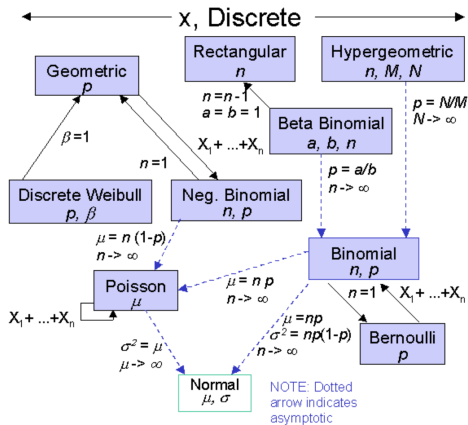


Figure: <https://statistical-engineering.com/relationships/>

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